



Sliding Bearing Design for the Pinion Shaft of High-speed Gearbox

Project Group ITR 3

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Interdisciplinary Engineering Project

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■ Problem Statement

- Define operating conditions and design limits for the hydrodynamic bearings for high-speed turbo gearbox pinion shaft using ISO VG 32 oil (speed, load, minimum film thickness, temperature, flow, and power-loss limits).
- Design and evaluate tilting pad, offset-halves, and three-lobe hydrodynamic bearings against these limits, comparing film thickness, temperatures, oil flow, power losses, stability, and cost for the selection of the safest and most efficient bearing for long term solution

■ Objective

- Design a hydrodynamic sliding bearing for the pinion shaft of a high-speed turbo gearbox operating with ISO VG 32 mineral oil
- Ensure reliable load-carrying capacity and hydrodynamic lubrication over the specified speed and load rang
- Limit frictional power losses, lubricant flow rate, and operating temperatures to meet efficiency and thermal constraints
- Verify that minimum oil film thickness and temperature limits at the bearing and sensor location are satisfied under nominal operating conditions

Operating Conditions

Condition	Requirement / Value
Maximum lubricant flow rate	$\leq 70 \text{ L/min}$
Maximum power loss (at max speed)	$\leq 40 \text{ kW}$
Minimum oil film thickness (nominal operation)	$\geq 20 \text{ }\mu\text{m}$
Maximum bearing temperature	$\leq 117 \text{ }^{\circ}\text{C}$
Temperature sensor location	Center plane of bearing, 5 mm below sliding surface
Sensor angular position	30° in direction of rotation from load direction
Maximum allowable sensor temperature	$\leq 99 \text{ }^{\circ}\text{C}$
Oil supply temperature	$40 \text{ }^{\circ}\text{C}$
Bearing force	$16.2 - 32.4 \text{ kN}$
Shaft speed	$5 - 15 \text{ krpm}$
Lubricant	ISO VG 32 mineral oil



Figure 1: Tilting 5 Pad Bearing

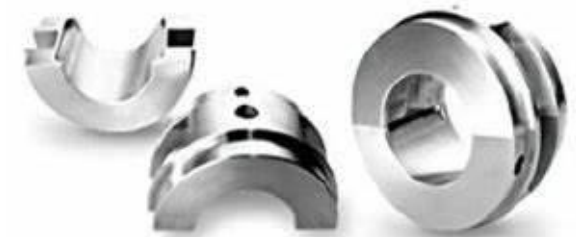


Figure 2: Offset Halves Bearing

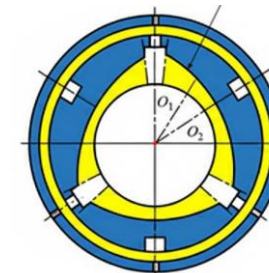


Figure 3: Three Lobe Bearing



Bearing Alternatives

1. Tilting Pad – 5 Pad Bearing

- **Bearing Design and Methodology**

- Bearing geometry is tuned by adjusting mounting angle, pivot offset, preload, and clearance.
- A 5-pad tilting-pad bearing is defined using pad arc, pivot position, preload, and clearance based on industrial practice.
- Load and rotational speed are kept constant across cases for fair comparison.
- Pad-by-pad hydrodynamic analysis is performed to compute pressure, flow, and load sharing.
- Minimum film thickness and temperatures are evaluated to ensure full-film lubrication and compliance with limits.
- Friction and power loss are evaluated, and geometry optimized to balance film thickness, temperature, and efficiency.

- **Selection Justification :**

- High-performance
- Turbomachinery standard
- Low cross-coupled stiffness
- Improved rotor stability
- Good stability-load-manufacturability compromise
- Avoids high-speed instability.

2.Offset Halves Bearing

- **Bearing Design and Methodology**

- Define bearing parameters – By setting diametral and relative clearances, balancing stability, flow, temperature, and power-loss targets.
- Geometry and design tuning – Iteratively adjust clearance, preload, and other geometric parameters to achieve desired performance.
- Iterative convergence loop – Use a pressure–temperature–viscosity loop to update film thickness, flow rate, metal/oil temperatures, and power loss until results converge.
- Hydrodynamic and thermo-elastohydrodynamic analysis – Compute pressure, flow, film thickness, and shaft/pad deformation under operating conditions.
- Performance evaluation – Check minimum film thickness, temperature limits, flow rate, and power loss against targets and safety margins.
- Design validation – Confirm the final design meets lubrication, thermal, and efficiency criteria, balancing stability and friction.

- **Selection Justification**

- Simple design
- Robust
- Easy manufacturing
- Good load capacity
- Moderate complexity
- Baseline/reference bearing



3.Three-Lobe Bearing

- **Bearing Design and Methodology**

- Define bearing geometry and parameters –Select a three-lobe bearing with three segments and a converging–diverging film.Specify lobe offset/eccentricity and diametral and relative clearance
- Geometry and clearance tuning – Iteratively adjust lobe offset, clearance, and segment geometry to optimize film formation and performance.
- Hydrodynamic analysis – Compute pressure distribution, flow, and film thickness in each lobe segment under operating conditions.
- Thermal and power assessment – Evaluate pad/shaft temperatures and frictional power loss to ensure compliance with limits and targets.
- Set performance targets – Establish flow rate, temperature limits, minimum film thickness, and power-loss targets for design evaluation.
- Sensitivity and validation study – Analyze how lobe offset and clearance variations affect film thickness, temperature, and power loss.

- **Selection Justification**

- Improved stability
- Directional stiffness
- Better than plain bearing
- Simpler than tilting-pad
- “Middle-ground” solution
- Lower cost/complexity

Comparative Presentation of Calculation Results

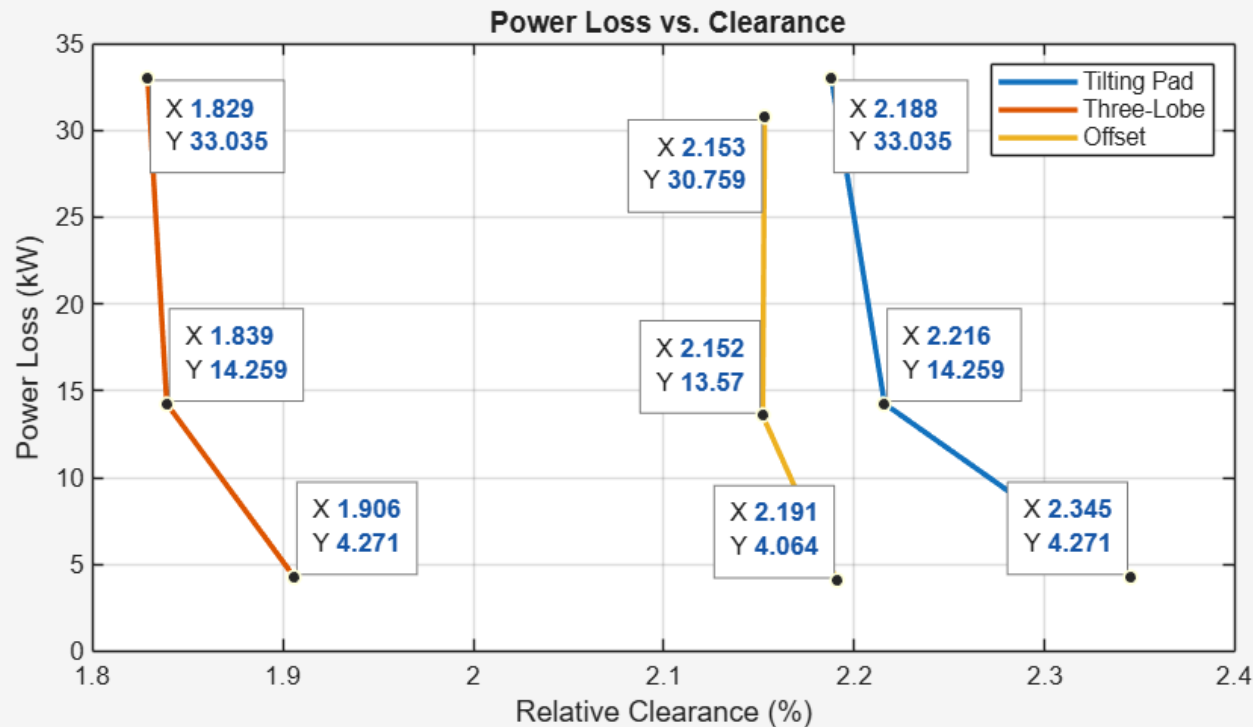


Figure 4: Power Loss vs Relative Clearance

Source : MATLAB

1. General Trend

Higher clearance → lower viscous shear
→ lower power loss.

2. Tilting 5-Pad Bearing

High loss at low clearance → drops fast
→ best efficiency at moderate clearance.

3. Three-Lobe Bearing

Similar trend → slightly lower loss than tilting pad
→ good efficiency–load balance.

4. Offset Halves Bearing

Power loss reduces sharply with clearance
→ lowest loss at high clearance but lower stability.

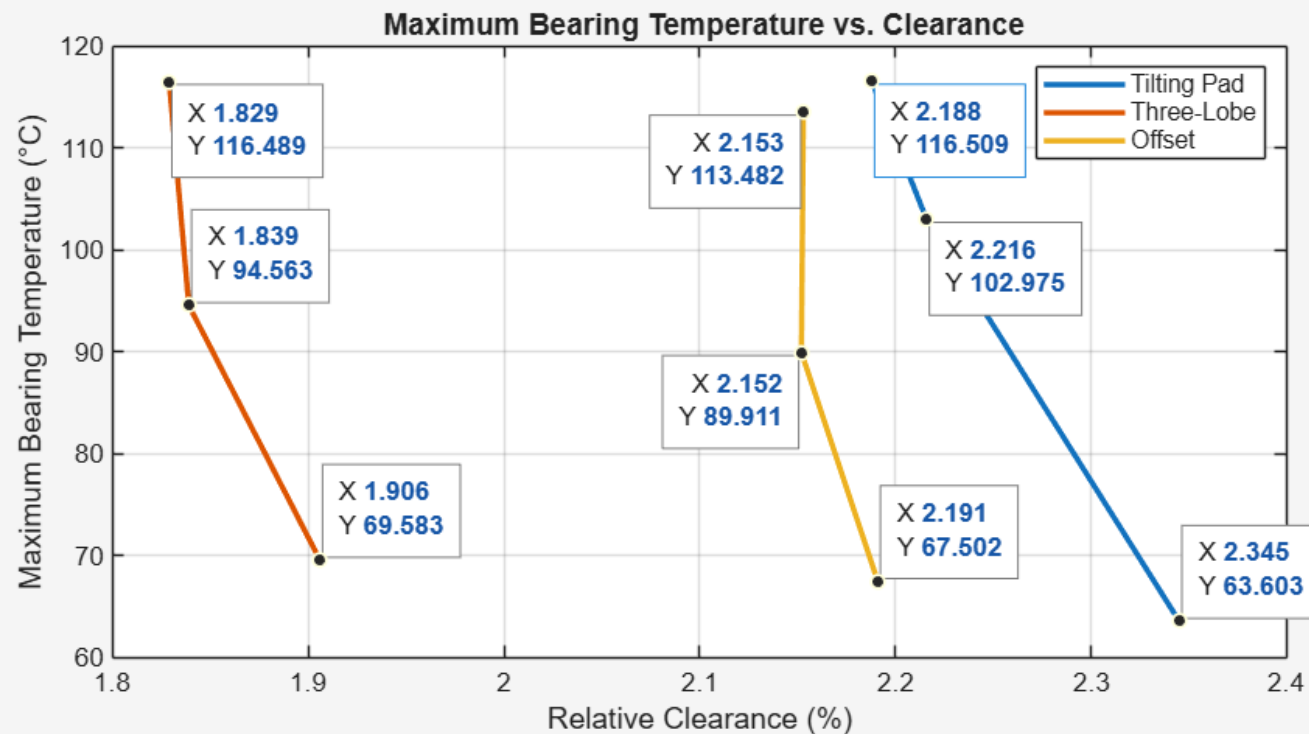


Figure 5 : Maximum Bearing Temperature vs Relative Clearance

Source : MATLAB

1. General Trend

Higher clearance → better oil circulation
→ lower bearing temperature.

2. Tilting 5-Pad Bearing

High temperature at low clearance →
drops significantly with clearance → best
thermal stability.

3. Three-Lobe Bearing

Similar temperature reduction trend →
slightly lower temperature than tilting pad
at some clearances.

4. Offset Halves Bearing

Temperature drops quickly with clearance
→ but more sensitive and less thermally
forgiving.

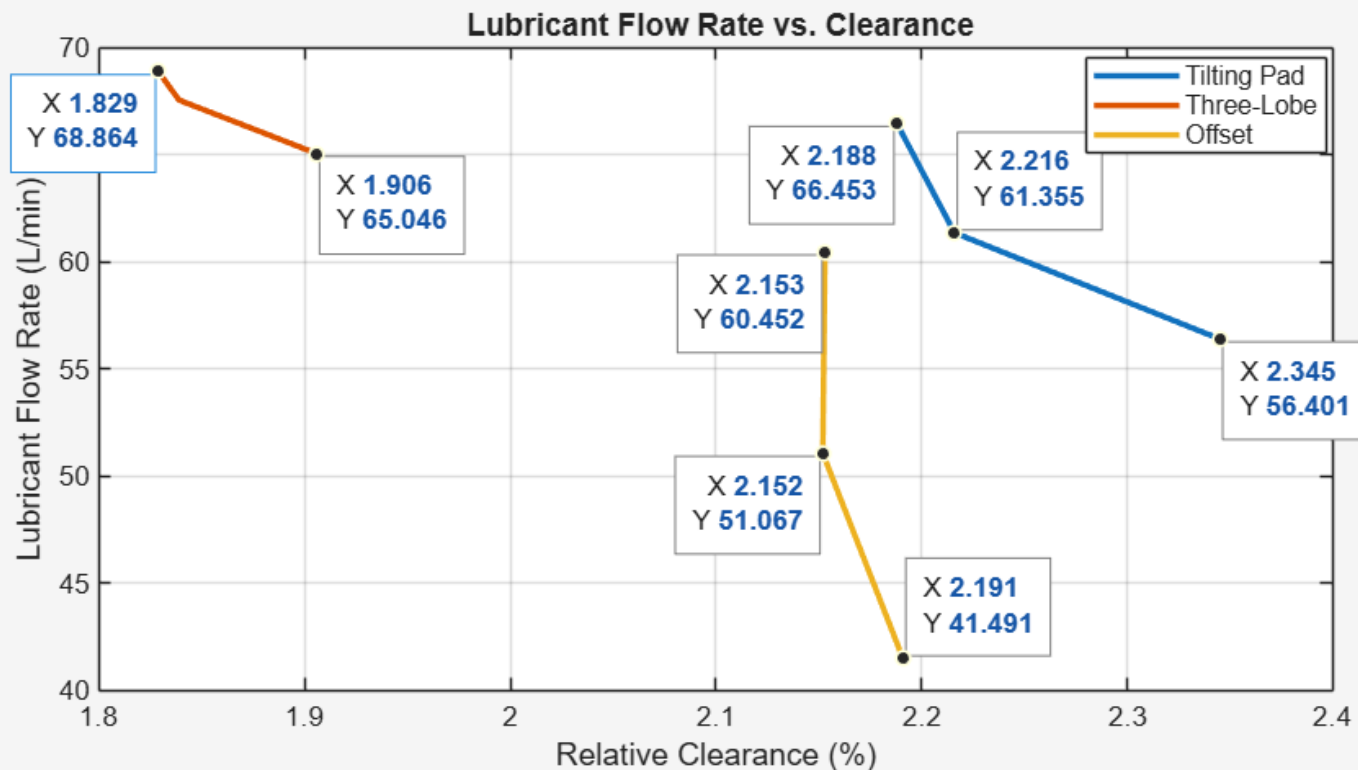


Figure 6 : Lubricant Flow Rate vs Relative Clearance

Source : MATLAB

1. General Trend

Higher clearance → lower hydrodynamic pressure, possible higher leakage.

2. Tilting 5-Pad

Bearing Flow decreases slowly with clearance → very stable lubrication.

3. Three-Lobe Bearing

High flow at low clearance → slightly reduces with clearance → good oil distribution.

4. Offset Halves

Bearing Flow drops sharply with small clearance increase → highly clearance sensitive

Summary of the Final Results

Parameter	Tilting Pad – 5 Pad		Offset Halves		Three Lobe		Operating Condition Requirement
Force (kN)	16.2	32.4	16.2	32.4	16.2	32.4	-
Speed (rpm)	5000	15000	5000	15000	5000	15000	-
Bearing Clearance (μm)	280.755	265.228	262.311	257.611	228.301	219.066	-
Lubricant Flow Rate (l/min)	56.401	66.453	41.491	60.452	65.046	68.864	≤ 70 l/min
Minimum Oil Film Thickness (μm)	31.518	22.380	43.988	38.659	43.005	38.121	≥ 20 μm
Sensor Temperature (°C)	31.86	36.84	46.04	59.74	40.07	53.76	≤ 99 °C
Max. Bearing Temperature (°C)	63.603	116.509	67.502	113.482	69.583	116.489	≤ 117 °C
Friction / Power Loss (kW)	4.271	33.035	4.061	30.759	4.271	33.035	≤ 40 kW

Conclusion and Concept Selection

Aspect	Tilting Pad – 5 Pad	Offset Halves	Three-Lobe
Lubricant flow rate & film thickness	Moderate flow rate , film thickness near lower limit at high load.	Lowest flow rate, thickest film at high load.	High flow rate, film thickness is moderate but needs more oil.
Bearing temperatures	Safest: stays within limits with good margin.	Within limits, slightly warmer than tilting pad.	Slightly exceeds max temperature at high load.
Power loss	Highest loss at high load (least efficient).	Lowest loss at high load (most efficient).	Intermediate loss.
Reliability	Best purely for reliability and thermal safety.	Good reliability; excellent film safety.	Limited by high temperature.
Cost & efficiency	Higher life-cycle cost (energy + oil).	Best overall cost and efficiency.	Medium cost, extra cooling/monitoring likely.

- Best overall choice: **Offset Halves Bearing**, as it offers the lowest power loss, best cost-efficiency, and sufficient reliability under typical operating conditions